

Full Stack Development with ServiceNow

Project Documentation

1. Introduction

- **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- **Team Members:**

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2. Project Overview

- **Purpose:** The project automates the procurement and configuration workflow for standard laptops in ServiceNow. Its objective is to ensure timely configuration after request approval, reduce manual intervention and errors, improve IT resource utilisation, and enhance procurement efficiency and productivity.
- **Features:** Service Catalog request for a Standard Laptop; Service Catalog trigger in Flow Designer; automatic creation of a Catalog Task after approval; automatic assignment to the Hardware group; predefined short description and task description; Approved status handling; visibility through Requested Item and Catalog Task records.

3. Architecture

- **Service Catalog / User Layer:** Users place a Standard Laptop request through the Service Catalog. The request generates a service request and Requested Item record.
- **Flow Designer:** A flow named “Standard Laptop Task” is configured to run as a System User. A Service Catalog trigger starts the flow when the relevant request conditions are met.
- **Task Automation:** The flow uses the ServiceNow Core “Create Catalog Task” action. The Requested Item record is passed into the request item field, while the task is populated with the short description and description “Laptop need to Configured”, assigned to the Hardware group, and associated with Approved status.
- **Hardware Team:** The generated Catalog Task is routed to the Hardware assignment group for laptop configuration.
- **Data / Records:** The workflow uses Service Catalog, Request, Requested Item, Approval, and Catalog Task records. The source project does not specify an external database; the workflow is implemented within ServiceNow.

4. Setup Instructions

- **Prerequisites:** A ServiceNow environment with access to Service Catalog, Flow Designer, Maintain Items, and the relevant Hardware assignment group.
- **Flow Setup:** Open ServiceNow → All → Flow Designer → New → Flow. Create “Standard Laptop Task”, select the Global application and System User execution, add the Service Catalog trigger, configure the Create Catalog Task action, save, and activate the flow.
- **Catalog Setup:** Open Maintain Items, select Standard Laptop, open Process Engine, remove the remaining automations if applicable, and assign the “Standard Laptop Task” flow to the catalog item.

5. Folder Structure

- **Application Structure:** The source project is a ServiceNow configuration/workflow implementation rather than a conventional React/Node folder-based application. Its main logical components are Service Catalog, Flow Designer flow, Request/Requested Item records, Approval records, and Catalog Task records.
- **Flow Components:** Trigger → Create Catalog Task action → task field values → Hardware assignment → approval-based execution.

6. Running the Application

- **User Workflow:** Open ServiceNow → Service Catalog → Hardware → Standard Laptop → Order Now. After the request is submitted and approved, the configured flow creates the Catalog Task automatically.
- **Verification:** Open the request number, review the Approvers section, approve the request, open the Requested Item, then open the Catalog Tasks section to verify the generated task, short description, and Hardware assignment.

7. API Documentation

- **ServiceNow APIs:** No custom REST API endpoints are specified in the source document. The solution uses ServiceNow's built-in Service Catalog and Flow Designer capabilities and the built-in Create Catalog Task action.
- **Workflow Interface:** The primary integration is the Service Catalog trigger and the ServiceNow Core Create Catalog Task action rather than a custom API.

8. Authentication

- **Access Control:** The source material does not define a custom authentication mechanism. Access is provided through the ServiceNow platform. The flow itself is configured to run as "System User" so that the automated task creation can execute independently of the individual requester.

9. User Interface

- **Service Catalog:** The user interface provides the Standard Laptop catalog item and an Order Now action for submitting the request.
- **Request Tracking:** Users/IT staff can view the Order Status, Request Number, Requested Item, Approvers, and Catalog Tasks records. The source document includes screenshots showing these ServiceNow screens and the resulting Hardware-assigned task.

10. Testing

- **Functional Test:** Submit a Standard Laptop request, verify that the request and approval record are created, approve the request, and confirm that the flow creates a Catalog Task.
- **Task Validation:** Verify that the generated task contains the expected short description, description, Approved status, and Hardware assignment group.
- **End-to-End Validation:** Confirm that the Hardware team can see and process the generated configuration task after approval.

11. Screenshots or Demo

- **Demo Flow:** The source documentation demonstrates Flow Designer creation and activation, Standard Laptop catalog configuration, request submission, approval, Requested Item inspection, and Catalog Task creation/assignment.
- **Key Screens:** Flow properties; Service Catalog trigger; Create Catalog Task action; Standard Laptop Process Engine configuration; Order Status; Approval record; Requested Item and Catalog Tasks record.

12. Known Issues

- **Manual Dependencies:** The source identifies the original process as having manual overhead and a risk of oversight or delay before automation.
- **Scope:** The documented workflow is focused on standard laptop procurement and configuration. Other hardware or non-standard requests are not described as implemented.
- **Missing Project Details:** Team member information and any custom integration/API specifications are not provided in the source material.

13. Future Enhancements

- **Broader Hardware Automation:** Extend the workflow pattern to additional IT hardware and service requests.
 - **Advanced Routing:** Introduce conditional routing based on laptop type, department, location, priority, or other procurement requirements.
- **Notifications:** Add automated notifications to requesters and hardware staff for approvals, task creation, configuration progress, and completion.
- **Reporting:** Add dashboards and metrics for request volume, approval time, configuration turnaround time, pending tasks, and resource utilisation.
- **Workflow Expansion:** Add additional Flow Designer actions and conditions as procurement requirements evolve.

Project outcome: automated standard laptop procurement and configuration workflow using ServiceNow Flow Designer.